# Enhanced Gateway Card Present REST API

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# Enhanced Gateway

## Card-Present Transactions

This short document outlines the transaction types, methods and the REST API offered by Enhanced Gateway (EG) to process credit card transactions in a card-present environment.

## Transaction Types:

EG REST API provides support for the following transaction types:

The endpoint to receive transaction request is:

**{{eg-card-present-endpoint}} = https://postransactions.com/connect/transact**

*A- Credit Card Transactions*
- *Sale : creditsale()*
- *Auth: creditauth()*
- *Capture (PostAuth): creditcapture()*
- *Void: creditvoid()*
- *Return: creditreturn()*
- *Adjust: creditadjust()*
- *Tokeinze: gettoken()*

*B- Debit Card Transactions*
- *Debit sale: debitsale()*
- *Debit Return: debitreturn()*

*C- EBT Transactions*
- *EBT Foodstamp : foodstampsale()*
- *EBT cash: ebtsale()*
- *EBT Return: ebtreturn()*

*D- Gift Card Transactions*
- *Issue a gift card: giftissue()*
- *Gift Card balance: giftbalance()*
- *Redeem a gift card: giftredeem()*

*E- Batch*
- *Close batch: closebatch()*

All card-present transactions use GET calls to perform and send transactions to designated and active devices.

A valid and active EG account is required before you can make any API calls.

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For .Net Framework users, we have built an DLL, which you can either directly add to your project, or you can add it from NuGet gallery.

For also provide a JavaScript library for web developments.

Please reach out to our sale engineers' team for more details.

# Credit Transactions

## Credit sale

A credit sale performs a sale transaction on a qualified, and active device on your account:

### Transaction parameters

| Parameter | Requirement | Description |
|---|---|---|
| **method** | Required | creditsale() |
| **account** | Required | Valid and active EG account number |
| **SN** | Required | Active Device serial number |
| **ticketid** | Required | A unique ticket ID or invoice number |
| **amount** | Required | Transaction amount |
| **Token** | Conditional – Required for Sale w/Token | A token representation of the card |
| **expirationdate** | Optional – Required for sale w/Token | Card's expiration date (mm/yyyy) |
| **tipreq** | Optional – Required to request tip at terminal | If set to 1, will ask the customer to select a tip amount |
| **tip** | Optional – Required if tip amount is sent along with the transaction amount | If set to an amount, the amount will be added to transaction as the tip amount |
| **merchant** | Optional – Required in Multi-Merchant scenarios | Send this parameter with merchant value of 1,…n to process a transaction in a multi-merchant system |

**Example**:

*{{eg-card-present-endpoint}}?method=creditsale&account=[Your EG accountNumber]&amount=2.50&ticketid=100&sn=[Your EG active device]*

**Response**:

{

"TransactionResult": "false",

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"PTK": "nD5pdVqKNm",

"ResponseCode": "100002",

"ApprovedAmount": "0.00",

"TransactionID": "83654813",

"AuthCode": "000000",

"Token": null,

"CardType": "",

"AccountNum": null,

"ExpDate": null,

"TicketID": "100",

"ResponseMsg": "ABORTED",

"HostCode": null,

"Timestamp": "2022-12-07 18:46:39",

"AvsResponse": null,

"CardHolder": null,

"BusinessId": "xxxxxxxxxx",

"UniqueId": "6390df8f5deda",

"TransType": "",

"EntryMode": null,

"RefNum": "",

"SerialNo": "xxxxxxxxxx",

"AmountDue": null,

"TipAmount": null,

"SurchargeAmount": "0.00",

"CvResponse": null,

"HostResponse": null,

"ExtraBalance": 0,

"CardBin": null,

"Tc": null,

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"Tvr": null,

"Aid": null,

"Tsi": null,

"Atc": null,

"AppLab": null,

"AppPN": null,

"Iad": null,

"Arc": null,

"Cid": null,

"Cvm": null,

"BatchNo": "",

"UserId": null,

"Method": "CREDIT"

}

After you received the response, first check the value of TransactionResult field. If the value is true, your transaction has been successful, otherwise it has failed. In case of failure, get the value of the ResponseMsg for more details.

To comply with regulations, you need to use the EMV fields, If the chipped card was dipped (or inserted) or contactless, to build your receipt.

```
Terminal:     2290077355
ID:           81622753
Type:         SALE
Card Number:  0775 - VISA
Method:       CONTACTLESS
AUTH:         792575
Message:     OK
AID:          A0000000980840
TVR:          0000000000
TSI:          0000
Invoice:      1
Order ID:     1
Customer:     CARDHOLDER/VISA

        Amount:  $0.01
        Total:   $0.01
Important:Retain this copy for your records
```

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# Credit Auth

Auth, creditauth, allows you to run a transaction and authorize and amount from the card. An authorization is not final, until your capture the amount in a follow up transaction. Please see Capture.

## Transaction parameters

| Parameter | Requirement | Description |
|---|---|---|
| **method** | Required | creditauth() |
| **account** | Required | Valid and active EG account number |
| **SN** | Required | Active Device serial number |
| **ticketid** | Required | A unique ticket ID or invoice number |
| **amount** | Required | Transaction amount |
| **Token** | Conditional – Required for Sale w/Token | A token representation of the card |
| **expirationdate** | Optional – Required for sale w/Token | Card's expiration date (mm/yyyy) |
| **merchant** | Optional – Required in Multi-Merchant scenarios | Send this parameter with merchant value of 1,…n to process a transaction in a multi-merchant system |

Example:

*{{eg-card-present-endpoint}}?method=creditauth&account=xxxxxxxxxx&amount=2.50&ticketid=100&sn=xxxxxxxx*

**Response**:

{

"TransactionResult": "true",

"PTK": "fgrv7BPM4l",

"ResponseCode": "000000",

"ApprovedAmount": "0.50",

"TransactionID": "83657885",

"AuthCode": "069377",

"Token": null,

"CardType": "VISA",

"AccountNum": "3487",

"ExpDate": "0225",

"TicketID": "101",

"ResponseMsg": "OK",

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"HostCode": null,

"Timestamp": "2022-12-07 19:06:11",

"AvsResponse": null,

"CardHolder": "JOHN DOE",

"BusinessId": "xxxxxxxxxx",

"UniqueId": "6390e423321a6",

"TransType": "AUTH",

"EntryMode": "CHIP",

"RefNum": "17",

"SerialNo": "xxxxxxxx",

"AmountDue": "0.00",

"TipAmount": "0.00",

"SurchargeAmount": "0.00",

"CvResponse": null,

"HostResponse": null,

"ExtraBalance": 0,

"CardBin": null,

"Tc": null,

"Tvr": "8000008000",

"Aid": "A0000000031010",

"Tsi": "6800",

"Atc": null,

"AppLab": null,

"AppPN": null,

"Iad": "06061203212000",

"Arc": "Z3",

"Cid": null,

"Cvm": null,

"BatchNo": "1",

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"UserId": null,

"Method": "CREDIT"

}

Although a creditauth() transaction is very similar to a creditsale(), but you have to keep in mind that, an auth is not final until it's captured.

## Credit Capture

A creditcapture() transaction, accepts the transaction ID of a previously ran creditauth() transaction, and attempts to capture the authorized amount, or a lesser or greater amount and finalize the transaction.

### Transaction parameters

| Parameter | Requirement | Description |
|---|---|---|
| **method** | Required | creditcapture() |
| **account** | Required | Valid and active EG account number |
| **SN** | Required | Active Device serial number |
| **ticketid** | Required | A unique ticket ID or invoice number |
| **amount** | Required | Transaction ID of the preceding creditauth() transaction. |
| **transactionid** | Required | A token representation of the card |

**Example**:

In the below example, we will capture a new amount for the creditauth() transaction we performed above.

{{eg-card-present-endpoint}}?method=creditcapture&account=xxxxxxxxxx&amount=2.50&ticketid=100&sn=xxxxxxxx&transactionid=83657885

**Response**:

{

"TransactionResult": "true",

"PTK": "Np14SpVL0r",

"ResponseCode": "000000",

"ApprovedAmount": "0.65",

"TransactionID": "83659400",

"AuthCode": "000000",

"Token": null,

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"CardType": "VISA",

"AccountNum": "3487",

"ExpDate": "0225",

"TicketID": "100",

"ResponseMsg": "APPROVED OFFLINE",

"HostCode": null,

"Timestamp": "2022-12-07 19:16:08",

"AvsResponse": null,

"CardHolder": null,

"BusinessId": "xxxxxxxxxx",

"UniqueId": "6390e678b844c",

"TransType": "POSTAUTH",

"EntryMode": "MANUAL",

"RefNum": "18",

"SerialNo": "xxxxxxxxxx",

"AmountDue": "0.00",

"TipAmount": "0.00",

"SurchargeAmount": "0.00",

"CvResponse": null,

"HostResponse": null,

"ExtraBalance": 0,

"CardBin": null,

"Tc": null,

"Tvr": "8000008000",

"Aid": "A0000000031010",

"Tsi": "6800",

"Atc": null,

"AppLab": null,

"AppPN": null,

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"Iad": "06061203212000",

"Arc": "Z3",

"Cid": null,

"Cvm": null,

"BatchNo": "1",

"UserId": null,

"Method": "CREDIT"

}

Creditcapture() does not require the use to insert a card. It will use the transaction ID of the creditauth() to complete the transaction.

Once you capture the transaction, you can capture it again.

## Credit Void

Use this transaction to void (cancel and remove it from the current batch). Use this transaction to void a previously performed creditsale(), creditauth() or creditcapture().

### Transaction parameters

| Parameter | Requirement | Description |
|---|---|---|
| **method** | Required | Creditvoid() |
| **account** | Required | Valid and active EG account number |
| **SN** | Required | Active Device serial number |
| **ticketid** | Required | A unique ticket ID or invoice number |
| **transactionid** | Required | A token representation of the card |

**Example**:

*{{eg-card-present-endpoint}}?method=creditvoid&account=xxxxxxxxxx&transactionid=83659400&sn=xxxxxxxx&ticketid=100*

**Response**:

{

"TransactionResult": "true",

"PTK": "JfNAYzK1eC",

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"ResponseCode": "000000",

"ApprovedAmount": "0.65",

"TransactionID": "83661085",

"AuthCode": "000000",

"Token": null,

"CardType": "VISA",

"AccountNum": "3487",

"ExpDate": "0225",

"TicketID": "100",

"ResponseMsg": "APPROVED OFFLINE",

"HostCode": null,

"Timestamp": "2022-12-07 19:26:48",

"AvsResponse": null,

"CardHolder": null,

"BusinessId": "xxxxxxxxxx",

"UniqueId": "6390e8f80c2e0",

"TransType": "VOID_POST",

"EntryMode": "MANUAL",

"RefNum": "19",

"SerialNo": "xxxxxxxx",

"AmountDue": "0.00",

"TipAmount": "0.00",

"SurchargeAmount": "0.00",

"CvResponse": null,

"HostResponse": null,

"ExtraBalance": 0,

"CardBin": null,

"Tc": null,

"Tvr": "8000008000",

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"Aid": "A0000000031010",

"Tsi": "6800",

"Atc": null,

"AppLab": null,

"AppPN": null,

"Iad": "06061203212000",

"Arc": "Z3",

"Cid": null,

"Cvm": null,

"BatchNo": "1",

"UserId": null,

"Method": "CREDIT"

}

Keep in mind that, creditvoid() is effective while the batch is still open. Once a batch is closed and transactions have been sent for processing, you cannot use this transaction.

# Credit Return

When you need to issue a refund you use creditreturn() transaction.

## Transaction parameters

| Parameter | Requirement | Description |
|---|---|---|
| **method** | Required | Creditreturn() |
| **account** | Required | Valid and active EG account number |
| **SN** | Required | Active Device serial number |
| **ticketid** | Required | A unique ticket ID or invoice number |
| **transactionid** | Required | Transaction ID of the transaction that you want to Retrun |
| **Amount** | Required | Amount to be returned to the card |

Creditreturn() requires the transaction ID of the target transaction, which you are trying to refund.

Creditreturn() requires the physical card to be present or at least card numbers, and card expiration date are available.

**Example**:

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*{{eg-card-present-endpoint}}?method=creditreturn&account=xxxxxxxxxx&ticketid=100&sn=xxxxxxxx&transactionid=83682494&amount=0.10*

**Response**:

{

"TransactionResult": "true",

"PTK": "Fx5hPVHbeD",

"ResponseCode": "000000",

"ApprovedAmount": "0.10",

"TransactionID": "83684173",

"AuthCode": "029283",

"Token": null,

"CardType": "VISA",

"AccountNum": "3487",

"ExpDate": "0225",

"TicketID": "100",

"ResponseMsg": "OK",

"HostCode": null,

"Timestamp": "2022-12-07 21:45:33",

"AvsResponse": null,

"CardHolder": "CARDHOLDER\\/VISA",

"BusinessId": "xxxxxxxxxx",

"UniqueId": "6391097da5a28",

"TransType": "RETURN",

"EntryMode": "CONTACTLESS",

"RefNum": "22",

"SerialNo": "xxxxxxxx",

"AmountDue": "0.00",

"TipAmount": "0.00",

"SurchargeAmount": "0.00",

"CvResponse": null,

"HostResponse": null,

"ExtraBalance": 0,

"CardBin": null,

"Tc": null,

"Tvr": "0000000000",

"Aid": "A0000000031010",

"Tsi": "0000",

"Atc": null,

"AppLab": null,

"AppPN": null,

"Iad": "06061203A00000",

"Arc": null,

"Cid": null,

"Cvm": null,

"BatchNo": "1",

"UserId": null,

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"Method": "CREDIT"

}

If you don't send an amount, you will be asked to enter an amount.

## Credit Adjust

We use creditadjust() to adjust a previously performed creditsale() transaction and add tip to it.

### Transaction parameters

| Parameter | Requirement | Description |
|---|---|---|
| **method** | Required | Creditadjust() |
| **account** | Required | Valid and active EG account number |
| **SN** | Required | Active Device serial number |
| **ticketid** | Required | A unique ticket ID or invoice number |
| **transactionid** | Required | Transaction ID of the sale transaction |
| **tip** | Required | Tip amount to be added to the transaction |

**Example**:

*{{eg-card-present-endpoint}}?method=creditadjust&account=1665525990&tip=0.05&ticketid=100&sn=3A904431&transactionid=83693904*

**Response**:

{

"TransactionResult": "false",

"PTK": "uscmOeqeq8",

"ResponseCode": "100004",

"ApprovedAmount": "0.00",

"TransactionID": "83694018",

"AuthCode": "000000",

"Token": null,

"CardType": "",

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"AccountNum": null,

"ExpDate": null,

"TicketID": "100",

"ResponseMsg": "UNSUPPORT TRANS",

"HostCode": null,

"Timestamp": "2022-12-07 22:38:49",

"AvsResponse": null,

"CardHolder": null,

"BusinessId": "xxxxxxxxxx",

"UniqueId": "639115f971935",

"TransType": "",

"EntryMode": null,

"RefNum": "",

"SerialNo": "xxxxxxxx",

"AmountDue": null,

"TipAmount": null,

"SurchargeAmount": "0.00",

"CvResponse": null,

"HostResponse": null,

"ExtraBalance": 0,

"CardBin": null,

"Tc": null,

"Tvr": null,

"Aid": null,

"Tsi": null,

"Atc": null,

"AppLab": null,

"AppPN": null,

"Iad": null,

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"Arc": null,

"Cid": null,

"Cvm": null,

"BatchNo": "",

"UserId": null,

"Method": "CREDIT"

}

Creditadjust() transaction only works if you have a Restaurant, or hospitality setting as your industry setting on your device. In other words, you cannot adjust a sale transaction, if you are in a retail setting.

## Tokenize a card

You can call gettoken() transaction to tokenize a card and store it. If your device is properly programmed, and your payment processor has activated tokenization on your account, then each creditsale() transaction, returns a token as well.

Your device must support this feature.

**Example**:

*{{eg-card-present-endpoint}}?method=gettoken&account=xxxxxxxxxx&sn=xxxxxxxx*

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